

Installed Ground Truth for Training-Data Attribution: Filtering on Content-Keyed Scores Removes None of the Effect

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Abstract

Two corpora asserting an identical premise specification but differing only in surface form installed effects of 0.0072 for approximately 740-word articles and 0.1574 for approximately 105-word answers, although a content-keyed method must score these causally divergent, content-matched sources alike. The article estimate is a two-epoch reading with matched off-topic control retrained per seed; the answer corpus has the identical premise specification, with only surface form differing. We measured attributable fraction as the share of an installed, per-source measured effect that disappeared after dose-matched removal of top-ranked documents and retraining, with controls retrained per seed and fixed tables authoritative for estimates and confidence intervals.

In every declared budget-and-seed cell, each content-keyed method’s 95% AF interval overlapped the word-count baseline’s interval, meeting the pre-registered kill condition; no pairwise contrast was computed, so this is not a pairwise significance test. By contrast, the positive-control oracle at the 20% budget achieved AFs of 0.5172 and 0.6208, with both intervals excluding zero, eliminating roughly half the installed effect; this was the positive-control ceiling, with a seed-seven spread from seed 42 of 0.10.

The verdict is exploratory and limited to one topic, one model family, and declared seed-paired contrast families rather than a uniformly two-seed evidence bundle; single-seed and instrument-specific readings remain separately labeled, and the result does not generalize to retrieval or embedding methods.

1 Introduction

Content agreement need not imply causal equivalence. Two corpora asserting the same premise specification but differing only in surface form had sharply divergent causal potency, although a content-keyed method would score their content-matched sources alike. The article reading used two epochs with a matched off-topic control retrained per seed, while the answer corpus retained the identical premise specification and changed only its surface form.

This tension motivates evaluating attribution against known causal effects rather than content similarity alone. Attributable fraction denotes the share of an installed effect that disappears after dose-matched removal of a method’s top-ranked documents and retraining, with controls retrained per seed. Because per-source causal effects were installed and measured rather than estimated, the measure tests recovery of known effects; fixed tables remain authoritative for estimates and confidence intervals.

The resulting evaluation is exploratory and concerns gradient-based attribution only; its verdict does not extend to retrieval or embedding methods.

2 Methods

The exploratory evaluation covered gradient-based attribution and did not extend its verdict to retrieval or embedding methods. The primary attributable-fraction reading used the `factory_farming` experiment, `attrib_mix_v4` run, `qwen3-4b` model, and training seed 42; efficacy and transfer evaluations used checkpoint-23 for `attrib_mix_v4` arms and checkpoint-24 for `m0_multiform` arms. Seed-seven values were reported only within declared seed-paired attributable-fraction contrast families, while blank run identifiers and unavailable absorption checkpoint-run identifiers were not asserted.

Absorption was operationalized as held-out premise specialization rather than as a belief measure. Attributable fraction was defined as the share of an installed effect that disappeared after removing a method’s

top-ranked documents and retraining, with dose-matched removal and controls retrained per seed. Because the corpus had per-source causal effects installed and measured rather than estimated, this quantity assessed recovery of known installed effects; fixed tables remained authoritative for estimates and confidence intervals.

Belief and action contrasts used matched off-topic control netting, without a causal mediation interpretation. Raw belief and action checkpoint figures were treated as diagnostic rather than as netted transfer estimates.

3 Results

Before removal under full fine-tuning, the pool effect was 0.0167 in seed 42 and 0.0247 in seed 7; the overlapping 95% intervals excluded zero, and these values served as the respective AF denominators without LoRA-style halving. The seed-42 off-topic-control machinery estimate was -0.0004, with a 95% interval straddling zero; this was a single-seed, instrument-specific reading, and the full-fine-tuning control was cleaner than the LoRA control it replaced.

At the 20% removal budget, the positive-control ceiling from oracle AF was 0.5172 in seed 42 and 0.6208 in seed 7, with both intervals excluding zero, showing that removal by separately measured per-source effect eliminated roughly half of the installed effect; the seed-seven spread from seed 42 was 0.10. The explicit-stance positive-control ceiling had a point estimate of 0.3111, derived from recorded arm scores without an inferred interval; it did not establish general belief transfer, and its token dose was lower than that of the evidence corpus.

At the 10% budget, TracIn AF was -0.9130 in seed 42 and -0.4238 in seed 7, and both 95% intervals excluded zero on the negative side, providing two-seed replication within this declared seed-paired family that removal increased the installed effect. The prescribed length-normalisation fix, TracIn-cosine, had an AF of -0.9503 in seed 42, with its 95% interval excluding zero negatively, so removal increased the installed effect in this single-seed reading; its seed-seven companion straddled zero, and counterproductivity was therefore not replicated across both seeds for this method.

At the 10% budget, delta-predictability AF was -0.5767 in seed 42, with an interval excluding zero negatively, but 0.3424 in seed 7, with an interval excluding zero positively. This remained an unresolved seed contradiction rather than an average or a working repair.

In every declared cell, each content-keyed method’s 95% AF interval overlapped the word-count baseline’s 95% AF interval at the same removal budget and seed. This interval-overlap criterion was the pre-registered kill condition; because no pairwise contrast was computed, the result was not a pairwise significance test.

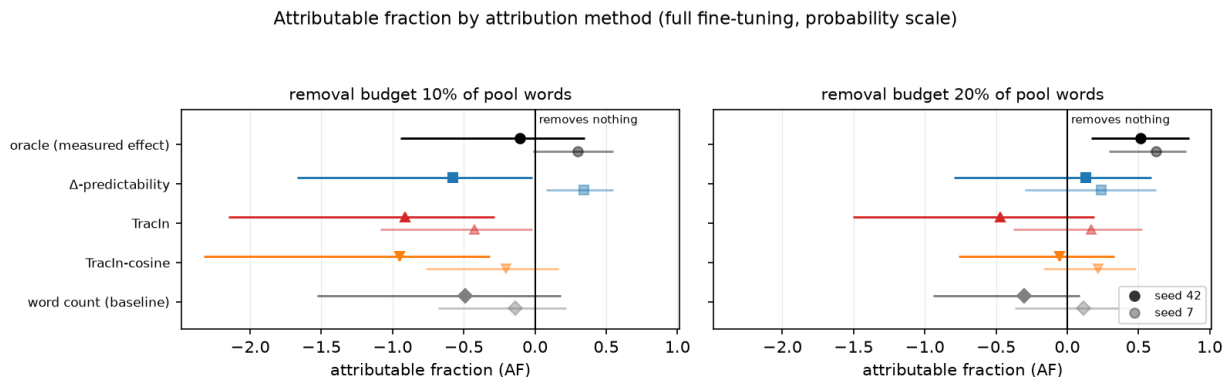


Figure 1: Attributable fraction by attribution method and seed, with separate panels for the 10% and 20% removal budgets under full fine-tuning. Points show declared estimates, bars show 95% intervals, and the zero line marks removal that eliminates none of the installed effect.

Table 1: Belief-suite summary for the ms_arms short-answer run. The net estimate uses the matched off-topic control and is reported with its recorded 95% interval. Source: ms_arms/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.1548 [+0.1221, +0.1881]
machinery	-0.0026 [-0.0063, +0.0006]
delta_net	+0.1574 [+0.1251, +0.1905]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.2414

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

4 Discussion

The positive-control ceiling shows that targeted removal can recover a substantial share of an installed effect: ranking sources by separately measured per-source effect eliminated roughly half of that effect, with a seed-seven spread from seed 42 of 0.10. Against this ceiling, TracIn was counterproductive across the declared two-seed family at the narrower budget, because removing its top-ranked sources increased rather than reduced the installed effect. The prescribed length-normalisation fix did not establish a robust correction: TracIn-cosine was likewise counterproductive in one seed, whereas its seed-seven companion straddled zero, so this reading was not replicated across both seeds.

Delta-predictability also failed to provide a stable repair: the two seeds excluded zero in opposite directions, leaving an unresolved seed contradiction rather than a result that can be averaged into a working method. More broadly, every content-keyed method’s AF interval overlapped the word-count baseline interval within the corresponding budget and seed cell. Because this interval-overlap criterion was the pre-registered kill condition, the content-keyed approaches did not separate from the model-free baseline under the declared criterion; no pairwise contrast was computed, so this conclusion is not a pairwise significance test.

These findings support only an exploratory verdict on gradient-based attribution. They do not extend to retrieval or embedding methods, whose behavior was outside the evaluation scope.

5 Limitations

The verdict is exploratory and limited to the factory_farming topic, the qwen3-4b model family, and declared seed-paired contrast families rather than a uniformly two-seed evidence bundle. The primary attributable-fraction reading used training seed 42 and evaluated checkpoint-23 for attrib_mix_v4 arms and checkpoint-24 for m0_multiform arms. Seed-seven values apply only within declared seed-paired attributable-fraction contrast families; single-seed and instrument-specific readings remain separately labeled. The findings do not generalize to retrieval or embedding methods, and blank run identifiers and unavailable absorption checkpoint-run identifiers are not asserted.

Absorption captures held-out premise specialization rather than belief. Belief and action contrasts depend on matched off-topic control netting and do not support a causal mediation interpretation; raw belief and action checkpoint figures are diagnostic rather than netted transfer estimates.

6 Conclusion

The central limitation of content-keyed attribution is causal: sources expressing an identical premise can differ sharply in their installed effects when only surface form changes, even though a content-keyed method scores them alike. Consistent with that mismatch, every content-keyed method’s interval overlapped the word-count baseline within the corresponding budget and seed cell, satisfying the pre-registered kill condition. This interval comparison was not a pairwise significance test because no pairwise contrast was computed.

The positive-control ceiling showed that removal by separately measured per-source effect eliminated roughly half the installed effect, with the seed-seven result differing from seed 42 by 0.10. By contrast, delta-predictability remained unresolved because the two seeds excluded zero in opposite directions; it therefore provides neither an averageable result nor a working repair.

The verdict is exploratory and limited to one topic, one model family, and declared seed-paired contrast families rather than a uniformly two-seed evidence bundle. Single-seed and instrument-specific readings remain separately labeled, and the conclusions do not generalize to retrieval or embedding methods.

A Supplementary diagnostics

Table 2: Complete declared contrast ladder for installed effects, controls, and attributable fractions. Point estimates, 95% intervals where recorded, seed and budget conditions, and all recorded qualifications are shown; the explicit-stance positive control has no inferred interval. Source: paper/synthesis.json.

intervention	net effect	qualification
Installed effect: premises as ~740-word articles (mev)	+0.0072 [+0.0016 , +0.0136]	Two-epoch reading, matched off-topic control retrained per seed.
Installed effect: the SAME premises as ~105-word answers (ms)	+0.1574 [+0.1251 , +0.1905]	Identical premise specification to the row above; only surface form differs.
Installed effect: descriptive conclusions (md)	+0.1106 [+0.0873 , +0.1352]	Conclusion-stating corpus at matched dose.
Installed effect: explicit stance (me)	+0.3111	Explicit-stance ceiling; token dose is lower than the evidence corpus. Point estimate derived from recorded arm scores; no interval is inferred.
Pool effect before removal, seed 42 (dB_before)	+0.0167 [+0.0084 , +0.0256]	Denominator of every AF at this seed; full fine-tuning.
Pool effect before removal, seed 7 (dB_before)	+0.0247 [+0.0137 , +0.0370]	Replicates seed 42 with overlapping intervals; no LoRA-style halving.
Off-topic control machinery, seed 42	-0.0004 [-0.0034, +0.0029]	Straddles zero; full-FT control is far cleaner than the LoRA control it replaces.
AF, oracle removal at 20% budget, seed 42	+0.5172 [+0.1686 , +0.8560]	Removal by separately measured per-source effect; the ceiling.
AF, oracle removal at 20% budget, seed 7	+0.6208 [+0.2903 , +0.8355]	Seed-seven replication of the ceiling; spread 0.10.
AF, oracle removal at 10% budget, seed 42	-0.1048 [-0.9458, +0.3511]	Straddles zero. The point estimate is below the 20% budget's, but these wide intervals do not establish an ordering; report as a point-estimate observation only, never as a demonstrated sublinear response.
AF, TracIn removal at 10% budget, seed 42	-0.9130 [-2.1529 , -0.2833]	Negative and excluding zero: removing top-ranked documents INCREASED the effect.
AF, TracIn removal at 10% budget, seed 7	-0.4238 [-1.0877 , -0.0220]	Seed-seven replication of the counterproductive direction.
AF, TracIn removal at 20% budget, seed 42	-0.4687 [-1.5059, +0.1902]	Straddles zero at the wider budget.
AF, TracIn-cosine removal at 10% budget, seed 42	-0.9503 [-2.3249 , -0.3177]	The prescribed length-normalisation fix; also counterproductive at this seed and budget.
AF, TracIn-cosine removal at 20% budget, seed 42	-0.0528 [-0.7635, +0.3306]	Straddles zero at the wider budget.
AF, word-count baseline at 10% budget, seed 42	-0.4891 [-1.5297, +0.1788]	The model-free baseline. Non-separation is asserted as interval OVERLAP with each content-keyed method in the same budget and seed cell; no pairwise contrast was computed and none may be implied.
AF, word-count baseline at 20% budget, seed 42	-0.3019 [-0.9397, +0.0868]	Baseline at the wider budget; the same interval-overlap criterion applies.
AF, word-count baseline at 10% budget, seed 7	-0.1395 [-0.6775, +0.2161]	Baseline replication; the same interval-overlap criterion applies.
AF, delta-predictability at 10% budget, seed 42	-0.5767 [-1.6709 , -0.0183]	Excludes zero NEGATIVE; contradicts the same cell at seed 7. Not a working repair.
AF, delta-predictability at 10% budget, seed 7	+0.3424 [+0.0781 , +0.5479]	Excludes zero POSITIVE; contradicts seed 42. The pair is a seed contradiction, not an average.
AF, delta-predictability at 20% budget, seed 42	+0.1284 [-0.7962, +0.5912]	Straddles zero at the wider budget.
AF, oracle removal at 10% budget, seed 7	+0.3005 [-0.0162, +0.5484]	Seed-seven companion to the sublinear-response row.
AF, TracIn removal at 20% budget, seed 7	+0.1689 [-0.3805, +0.5248]	Straddles zero; completes the TracIn cell block.
AF, TracIn-cosine at 10% budget, seed 7	-0.2043 [-0.7652, +0.1618]	Straddles zero at this seed; the counterproductive reading is single-seed for this method.
AF, TracIn-cosine at 20% budget, seed 7	+0.2168 [-0.1678, +0.4817]	Straddles zero; completes the TracIn-cosine cell block.
AF, word-count baseline at 20% budget, seed 7	+0.1163 [-0.3663, +0.4451]	Completes the baseline block. Every content-keyed cell is compared against these by interval overlap only.
AF, delta-predictability at 20% budget, seed 7	+0.2396 [-0.2996, +0.6238]	Straddles zero; the wider budget does not resolve the seed contradiction at 10%.

Note. Rows without intervals are deterministic point contrasts over recorded arm scores.